

NAVEGANTES, Brazil

WMO: 825991

Lat: 26.880S

Lon: 48.651W

Elev: 18

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	48.0	50.6	40.9	37.7	53.6	44.4	43.3	56.3	19.0	62.4	17.1	61.6	9.8	260	0.329

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.1	89.6	80.2	87.6	79.9	85.9	79.2	82.6	86.3	81.1	84.9	80.1	83.9	11.2	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
82.0	166.8	85.1	80.5	158.4	83.9	79.0	150.5	82.7	46.5	85.3	44.9	85.7	43.9	84.3	88.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.2	16.4	14.8	42.2	97.0	2.7	2.5	40.3	98.8	38.7	100.2	37.2	101.6	35.2	103.4		
(5)			41.5	84.5	2.9	1.9	39.4	85.9	37.7	87.0	36.1	88.1	34.0	89.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	71.4	79.2	79.1	76.9	73.7	69.0	64.7	63.3	64.5	67.1	70.4	73.1	76.4
	DBStd	6.81	3.19	3.15	3.15	4.37	4.08	4.87	4.86	4.84	3.95	3.48	3.21	3.85
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	261	0	0	0	4	12	63	87	70	23	2	0	0
	CDD50	7814	905	815	833	712	588	442	412	448	514	632	693	820
	CDD65	2603	440	396	368	266	135	55	34	53	88	170	243	355
	CDH74	15859	3961	3517	2661	1454	305	92	42	82	96	328	920	2401
	CDH80	3460	1151	936	514	202	25	11	6	15	6	30	64	500
Wind	WSAvg	7.7	8.8	8.1	8.0	7.1	6.9	6.7	6.5	7.1	7.6	8.0	9.2	8.7
Precipitation	PrecAvg	70.0	10.2	7.9	7.2	3.7	4.2	4.3	5.0	3.5	5.8	6.7	5.4	6.2
	PrecMax	99.3	15.4	12.5	11.5	8.3	14.3	8.5	12.1	11.5	12.2	13.7	10.0	9.6
	PrecMin	56.7	5.9	2.8	2.2	1.4	0.6	2.1	0.8	0.5	0.7	2.9	2.9	1.9
	PrecStd	10.4	2.8	2.8	2.3	1.9	3.7	1.4	2.9	3.0	2.8	2.6	2.0	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	93.6	91.4	89.4	87.8	82.6	80.8	79.0	82.0	80.5	84.0	85.9	91.8
		MCWB	80.6	80.8	78.9	79.2	76.2	70.5	71.2	71.8	71.8	73.6	74.1	79.1
	2%	DB	89.6	88.2	86.1	84.3	79.0	76.7	73.6	75.4	76.7	79.1	81.0	87.4
		MCWB	80.8	81.3	78.8	78.8	73.0	69.3	67.1	69.2	71.4	73.2	74.8	79.4
	5%	DB	87.6	86.4	84.3	82.2	77.0	73.5	71.7	73.2	74.8	77.1	79.2	84.4
		MCWB	80.2	80.3	78.3	76.9	71.9	68.3	66.9	69.2	70.2	71.5	73.8	77.9
	10%	DB	85.8	84.6	82.6	80.5	75.3	71.5	69.8	70.2	73.1	75.4	78.6	82.5
		MCWB	78.9	79.1	76.7	75.5	71.0	67.2	66.2	66.8	69.0	70.3	73.3	76.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	83.9	84.1	82.6	84.4	77.5	73.5	72.0	73.7	75.3	79.0	80.7	82.6
		MCDB	88.8	88.0	84.8	85.8	81.2	78.1	76.3	75.8	77.1	79.1	80.7	87.6
	2%	WB	82.0	82.5	80.5	80.7	74.9	71.3	69.2	71.5	72.5	75.2	77.5	80.5
		MCDB	87.0	86.0	83.6	82.2	77.5	74.5	71.9	74.8	75.0	77.2	78.8	84.6
	5%	WB	80.5	81.1	79.2	78.7	73.5	69.6	67.7	69.6	71.1	73.2	76.0	79.1
		MCDB	85.6	85.1	82.4	80.3	76.0	72.2	70.4	72.0	73.4	75.5	77.9	83.0
	10%	WB	79.3	80.0	78.1	76.9	72.0	68.4	66.7	68.1	69.9	71.7	74.5	77.8
		MCDB	84.0	83.9	81.5	79.0	74.4	70.6	69.4	70.5	72.0	74.5	77.2	81.6
Mean Daily Temperature Range	5% DB	MDBR	12.1	10.9	11.3	12.0	11.5	12.4	12.0	11.8	10.1	9.1	9.9	10.5
		MCDBR	15.2	13.0	12.8	13.0	13.3	14.9	14.0	14.0	11.7	11.1	11.2	14.1
		MCWBR	8.3	7.3	7.4	7.8	8.0	8.8	8.6	8.4	7.1	6.9	6.9	7.7
	5% WB	MCDBR	13.5	11.9	12.1	11.0	11.6	12.4	12.3	12.6	10.3	9.6	10.1	12.5
MCWBR		8.4	7.4	8.0	8.4	8.0	8.4	8.4	8.3	7.3	7.2	7.6	7.9	
Clear-Sky Solar Irradiance	taub	0.415	0.408	0.397	0.390	0.371	0.390	0.393	0.460	0.523	0.464	0.424	0.419	
	taud	2.439	2.458	2.482	2.482	2.506	2.441	2.415	2.174	1.992	2.229	2.361	2.404	
	Ebn at Noon	295	292	285	272	263	248	252	245	244	272	290	294	
	Edh at Noon	39	37	35	32	28	29	30	42	56	46	42	40	
All-Sky Solar Radiation	RadAvg	1802	1712	1493	1231	955	820	880	1053	1180	1357	1657	1774	
	RadStd	135	148	145	116	104	66	105	108	149	194	169	159	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	-0.50	-0.79	N/A	N/A	N/A	N/A

Nomenclature: See separate page